Session Objectives

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- Explain the epidemiology, pathogenesis and transmission of TB
- Identify risk factors associated with TB
- Understand patient presentation and physical examination findings consistent with active TB
- Differentiate the radiologic findings of TB
- Differentiate between testing modalities for screening and diagnosis of active vs. latent TB

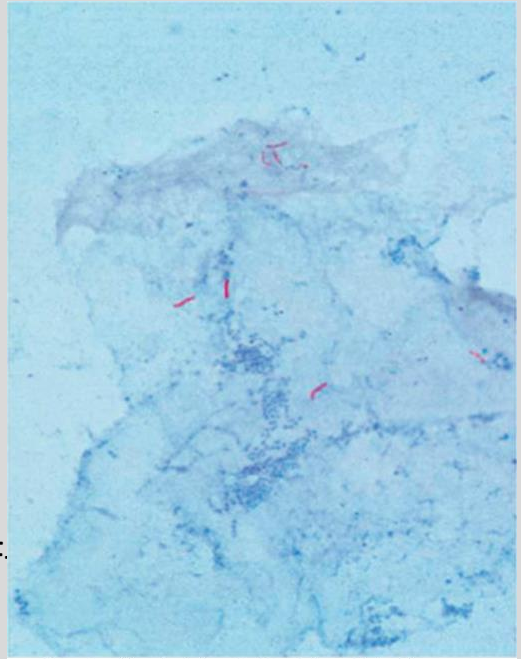
Source: Course Syllabus

Mycobacterium tuberculosis

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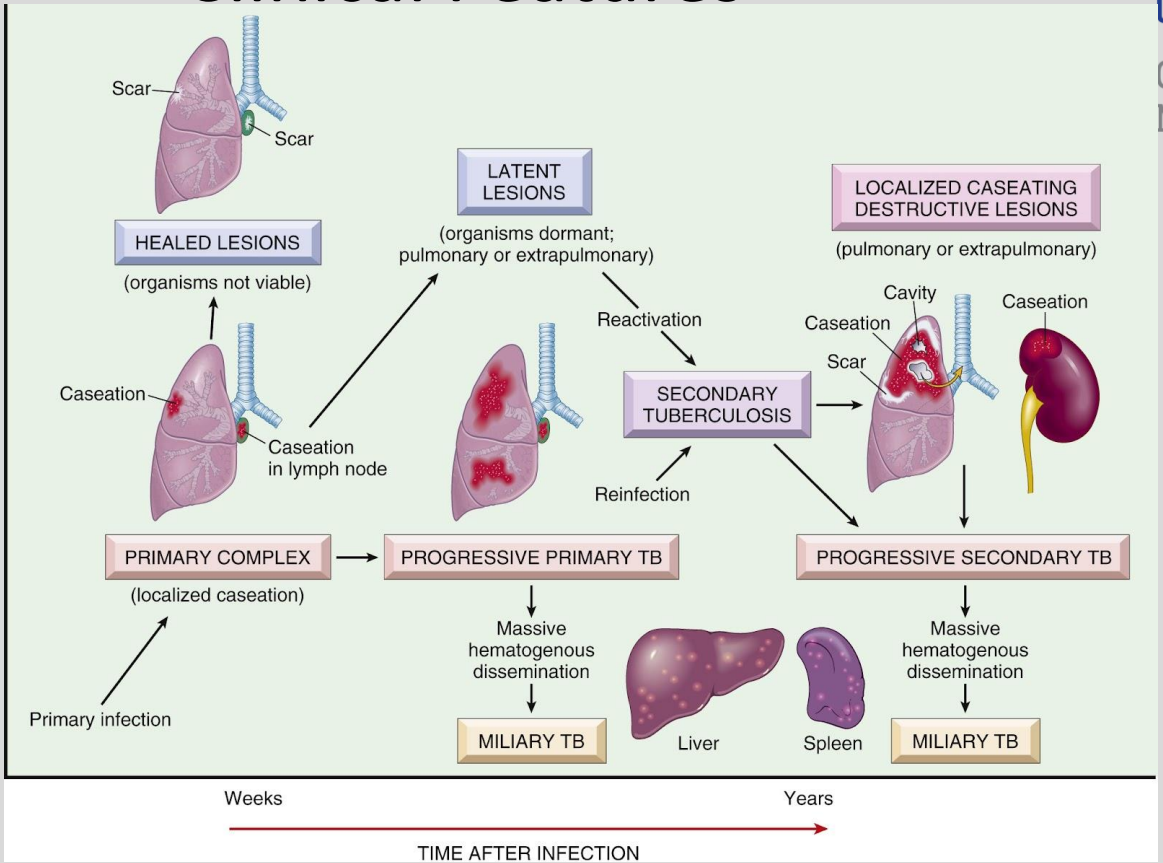
- Rod-shaped, non-spore forming, aerobic bacteria
- Facultative intracellular
- Acid Fast → high content of mycolic acid
  - Mycolic acid: long-chain cross linked fatty acid
- Neutral on gram stain
  - Mycolic acid makes impervious to staining
- Slow growing
- Virulence factors:
  - Cord factor: activates macrophages and induces release of TNF-alpha
  - Sulfatides: inhibit phagolysosome fusion



Source: Joseph Loscalzo, Anthony Fauci, Dennis Kasper, Stephen Hauser, Dan Longo, J. Larry Jameson: Harrison's Principles of Internal Medicine, 21e
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Harrison's Principles of Internal Medicine, 21e. Chapter 178: Tuberculosis.

Clinical Features



Source: Source: Robbins & Cotran Pathologic Basis of Disease, Chapter 8 Infectious Diseases.

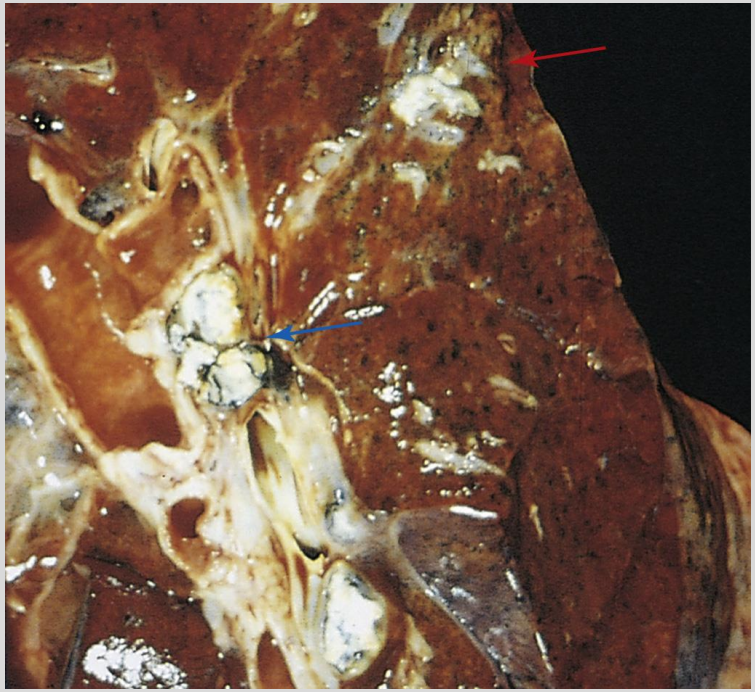
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Histopathology

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- Primary:
  - Ghon complex:
    - Ghon focus: 1-1.5 cm area of gray-white inflammation with consolidation (red arrow)
    - Middle and lower lung zones
    - Hilar Lymph Nodes with caseation (Blue arrow)
  - Ghon complex undergoes fibrosis → calcification → Ranke complex
  - Histology: Central caseation surrounded by epithelioid and multinucleated giant cells

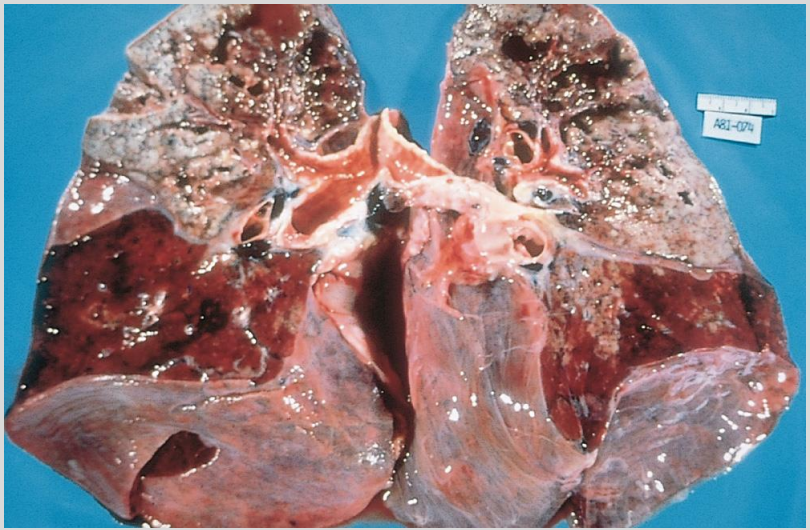


Source: Robbins & Cotran Pathologic Basis of Disease, Chapter 8 Infectious Diseases.

Histopathology

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- Secondary:
  - Lesion: small focus of consolidation, <2cm diameter, within 1-2cm of apical lung
  - Initial parenchymal focus undergoes progressive fibrous encapsulation → fibrocalcific scars
  - Histology: coalescent tubercles with central caseation



Source: Robbins & Cotran Pathologic Basis of Disease, Chapter 8 Infectious Diseases

Clinical Features: Extra-Pulmonary

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- Tuberculous lymphadenitis: TB of lymph nodes
- Pleural TB: an exudative pleural effusion with predominantly lymphocytes
- Upper airway TB: larynx, pharynx and epiglottis
- Genitourinary TB: symptoms may resemble UTI
- Skeletal TB: **Pott's disease**: TB of vertebral bodies
- Tuberculous meningitis
- Tuberculoma: Space occupying lesion in brain
- Gastrointestinal TB: terminal ileum, cecum
- Pericardial TB



Source: J.L. Jameson, A.S. Fauci, D.L. Kasper, S.L. Hauser, D.L. Longo, J. Loscalzo: Harrison's Principles of Internal Medicine, 20th Edition
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Diagnosis: Physical Exam

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- General:
  - Pallor, clubbing (advanced disease), generalized wasting
- Pulmonary:
  - May have no abnormalities on examination
  - Consolidation: dullness to percussion, crackles, diminished breath sounds
  - Cavitation: hyperresonance on percussion, amphoric breath sounds
  - Bronchial obstruction: rhonchi
  - Pleural effusion and/or empyema: dullness to percussion, diminished breath sounds
- Extra-pulmonary:
  - Depends on system involved



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Differential Diagnosis

- Pneumonia
- Malignancy
- Non-tuberculous mycobacterium
- Fungal infection
- Sarcoidosis

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Diagnosis: Testing

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Mantoux tuberculin skin test (TST)

- T-cell mediated delayed type hypersensitivity reaction
- Takes 2-8 weeks after initial infection for immune system to react to PPD
- Positive test: Person's body was infected with TB bacteria → need additional tests to determine if LTBI or TB disease
- Negative test: Person's body did not react to test
- False-negative: Anergy, low protein states, diseases affecting lymphoid organs
- False-positive: Infection w/ nontuberculous mycobacteria, previous BCG vaccination
- Absence of reaction to TST does NOT exclude diagnosis



https://www.cdc.gov/tb/index.html
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Diagnosis: Imaging

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Summary

- TB is a bacterial infection caused by *Mycobacterium tuberculosis.*
- It is spread via inhalation of droplet particles of an infected persons and mainly affects the lungs, but can affect any part of the body.
- Not everyone infected with TB bacteria becomes sick
- There are two TB-related conditions: latent TB infection and TB disease.
- While most patients will not become infectious, if active disease is left untreated, it is deadly.
- The clinical suspicion and diagnostic results are important when determining the patient's course of treatment.

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Knowledge Check

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Question#1: Which of the following groups are more likely to be exposed to or infected with M. tuberculosis?

1. Contacts of people known or presumed to have infectious TB
2. People who live in, or have lived in, high-risk congregate settings (for example, homeless shelters or correctional facilities)
3. Health care workers who serve patients with TB disease
4. People born in or who frequently travel to countries where TB disease is common, including Mexico, the Philippines, Vietnam, India, China, Haiti, and Guatemala

A. 1, 2, and 3 are correct
B. 1 and 3 are correct
C. 2 and 3 are correct
D. Only 4 is correct
E. All are correct

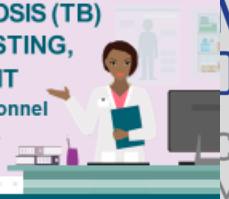
https://www.cdc.gov/tb/hcp/education/core-curriculum-on-tuberculosis.html

In May 2019, CDC and the National TB Controllers Association issued

TUBERCULOSIS TESTING + TREATMENT OF U.S. HEALTH CARE PERSONNEL

UPDATED RECOMMENDATIONS FOR TUBERCULOSIS (TB) SCREENING, TESTING, AND TREATMENT

for health care personnel in the United States.



What are the updated recommendations?

Before starting a new job in a health care setting, all workers and volunteers should receive

- TB individual risk assessment
- Symptom screening
- TB test



An annual TB test is not recommended unless there is a known exposure or ongoing transmission.

All health care personnel should receive TB education every year.

Treatment for latent TB infection (LTBI) is strongly encouraged for health care personnel diagnosed with LTBI.

Shorter treatment regimens should be used.



Who is affected by the new recommendations?

Individuals who work or volunteer in health care settings



Health care settings include

- Inpatient and outpatient settings
- Laboratories
- Emergency medical services
- Medical settings in prisons or jails
- Home-based health care settings
- Long-term care facilities





What if my state's regulations are different?

Follow your state's guidance.



For TB regulations in your area, please contact your state or local TB control program.



Where can I get more information? www.cdc.gov/tb

CDC Centers for Disease Control and Prevention National Center for HIV/AIDS, Viral Hepatitis, STD, and TB Prevention